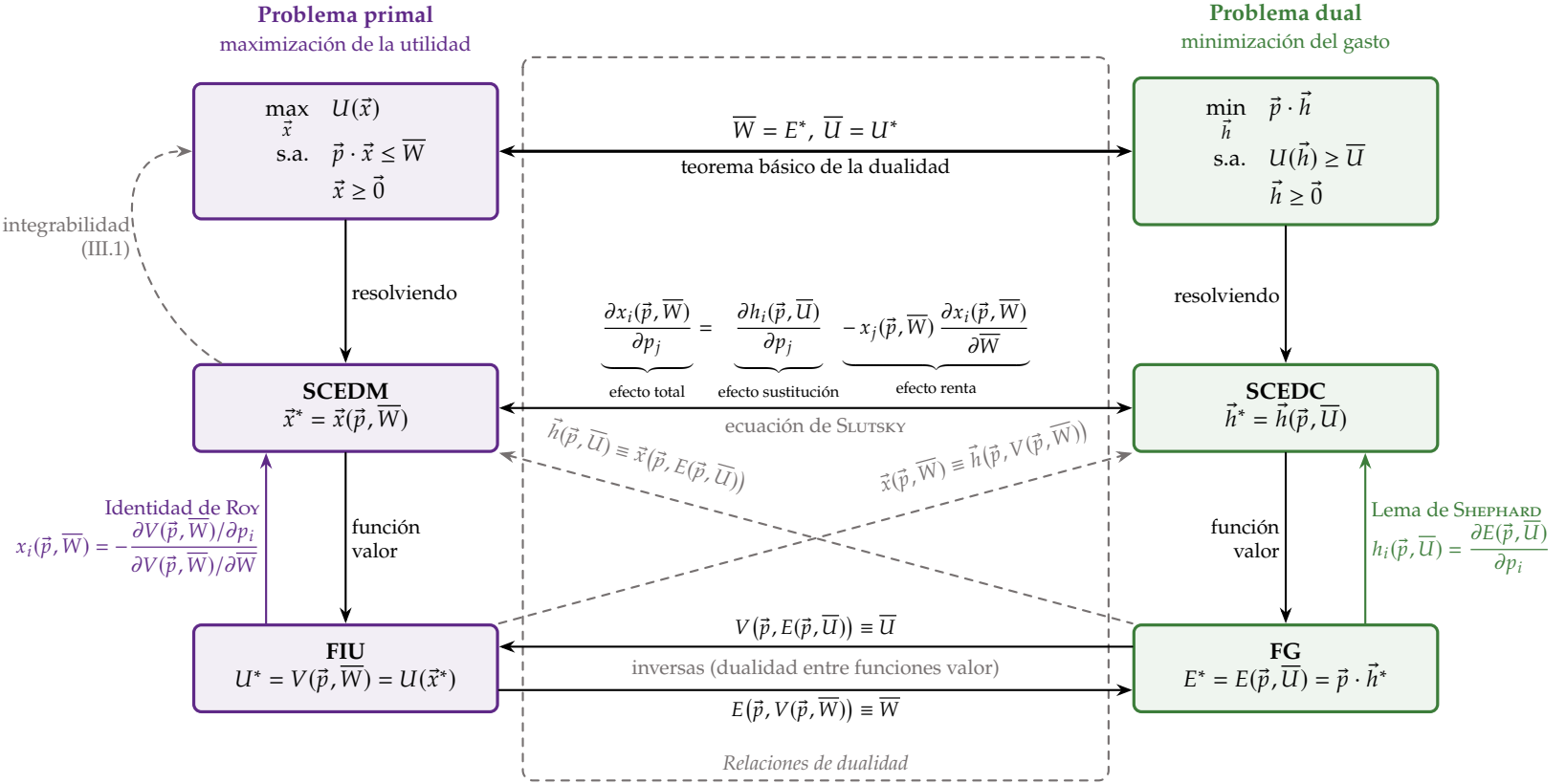

Problema primal
maximización de la utilidad
Problema dual
minimización del gasto
$\max_{\vec{x}} \; U(\vec{x})$
s.a. $\vec{p}\cdot\vec{x} \le \overline{W}$
$\vec{x} \ge \vec{0}$
$\min_{\vec{h}} \; \vec{p}\cdot\vec{h}$
s.a. $U(\vec{h}) \ge \overline{U}$
$\vec{h} \ge \vec{0}$
$\overline{W} = E^*,\ \overline{U} = U^*$
teorema básico de la dualidad
integrabilidad
(III.1)
resolviendo
resolviendo
$\frac{\partial x_i(\vec{p},\overline{W})}{\partial p_j} = \frac{\partial h_i(\vec{p},\overline{U})}{\partial p_j} - x_j(\vec{p},\overline{W})\frac{\partial x_i(\vec{p},\overline{W})}{\partial \overline{W}}$
efecto total
efecto sustitución
efecto renta
SCEDM
$\vec{x}^* = \vec{x}(\vec{p},\overline{W})$
ecuación de Slutsky
SCEDC
$\vec{h}^* = \vec{h}(\vec{p},\overline{U})$
$\vec{h}(\vec{p},\overline{U}) \equiv \vec{x}(\vec{p},E(\vec{p},\overline{U}))$
$\vec{x}(\vec{p},\overline{W}) \equiv \vec{h}(\vec{p},V(\vec{p},\overline{W}))$
Identidad de Roy
$x_i(\vec{p},\overline{W}) = -\frac{\partial V(\vec{p},\overline{W})/\partial p_i}{\partial V(\vec{p},\overline{W})/\partial \overline{W}}$
función
valor
función
valor
Lema de Shephard
$h_i(\vec{p},\overline{U}) = \frac{\partial E(\vec{p},\overline{U})}{\partial p_i}$
$V(\vec{p},E(\vec{p},\overline{U})) \equiv \overline{U}$
inversas (dualidad entre funciones valor)
$E(\vec{p},V(\vec{p},\overline{W})) \equiv \overline{W}$
FIU
$U^* = V(\vec{p},\overline{W}) = U(\vec{x}^*)$
FG
$E^* = E(\vec{p},\overline{U}) = \vec{p}\cdot\vec{h}^*$
Relaciones de dualidad